Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18ME55]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

CAD CAM

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	Explain the scope and applications of CAD/CAM.	08
	b.	Define wire frame model and solid model.	06
2.	a.	Explain Graphical Kernel System (GKS) and Initial Graphics Exchange Specification (IGES).	10
	b.	What is the need of Graphic Standard?	04
3.	a.	What is homogeneous transformation?	06
	b.	Explain translation, scaling, rotation, clipping.	08
		OR	
4.	a.	What is CNC architecture? State application advantages and disadvantages of CNC	10
	b.	What are encoders?	04
5.	a.	Explain with a sketch of automatic tool changer. State the advantages.	08
	b.	Explain re-circulating ball screws.	06
		OR	
6.	a.	Write part program for the part shown in figure 6a	08

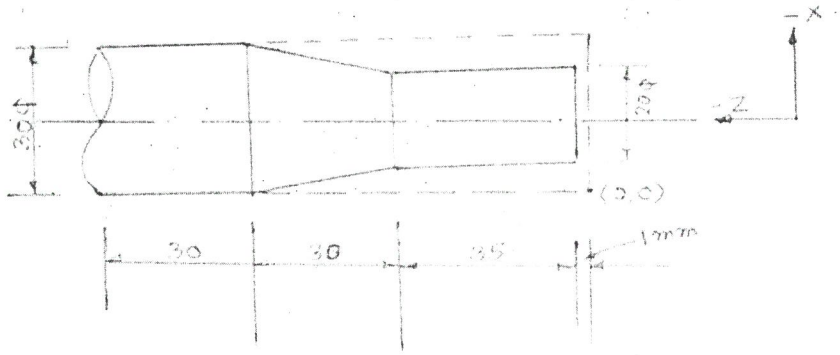

Fig. 6a

	b.	Explain the sequential number, preparatory codes and miscellaneous codes in part program.	06
7.	a.	What is generative process planning?	06
	b.	What is process planning? Name the steps involved in process planning?	08

OR

8.	**a.**	Discuss on the four basic robot configurations available commercially.	**08**
	b.	What are end effecters? List its types.	**06**